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2D material channel-based phototransistor array for broadband image sensing

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Abstract

The demand for photodetectors and image sensors has grown exponentially due to their applications in security surveillance, robotics, automotive systems, image recognition, and military technologies. Two-dimensional (2D) materials are being extensively explored for broadband-responsive photodetectors [1]. In this study, we report high-performance broadband phototransistors by leveraging the unique advantages of 2D materials, high-*k* dielectrics, and silicon. By combining photoconduction, photogating, and swift carrier extraction mechanisms in 2D materials, our devices - composed of 2D materials, HfO_2, and Si - exhibit high responsivity, external quantum efficiency, and detectivity across a broad spectrum from UV to near-infrared wavelengths, even at low operating bias. The fabricated device array demonstrates exceptional uniformity and reproducibility, highlighting its potential for practical applications.

1. Introduction

Advances in information science and artificial intelligence (AI), including visual recognition, surveillance systems, humanoid robotics, and defense technologies, are driving the need for high-performance sensing devices. To meet these demands, significant efforts have been directed toward the design of functional materials, structural modifications, and the miniaturization of sensing and imaging devices.

Current imaging technologies, primarily complementary metal-oxide semiconductor (CMOS) and charge-coupled device (CCD) systems, have achieved remarkable progress through innovations in architectural design and novel materials for channel layers and gate oxides. While CCDs offer high sensitivity and superior image quality with lower noise compared to CMOS sensors, further improvements are sought. Two-dimensional (2D) materials such as graphene, MoS_2, and $MoTe_2$, along with their heterostructures combined with conventional semiconductors like Si and Ge, have been widely investigated for photodetectors (PDs). These materials exhibit strong interlayer coupling, tunable energy band structures, and van der Waals surfaces, making them highly promising. However, phototransistors typically employ low-*k* SiO_2 as the dielectric layer, which necessitates high operating bias.

In this work, we present high-performance phototransistors that utilize 2D materials (e.g., graphene and $MoTe_2$) as the channel, combined with high-k HfO_2 and Si. The devices achieve superior photoresponse by integrating photoconduction (PC), photogating (PG), photovoltage effects, and electrostatic doping. These phototransistors exhibit broadband response from UV to near-infrared wavelengths, with high responsivity underscoring their potential for solar-blind UV photodetection and broadband image sensing.

2. Experimental methods

Synthesis of 2D materials

Graphene was synthesized by chemical vapor deposition method. The detailed processes can be found in our previous reports [2]. $MoTe_2$ layers were prepared by the thermally assisted conversion (TAC) technique. First, a Mo layer was deposited on the substrate using magnetron sputtering, followed by tellurization to form $MoTe_2$ film.

Fabrication of phototransistors

The phototransistors were fabricated by depositing a 10–20 nm HfO_2 layer on a Si substrate using atomic layer deposition (ALD) at 250 °C. Graphene was then transferred, or $MoTe_2$ was directly grown, onto the dielectric surface. Metal electrodes were patterned to establish contact with the 2D material layer.

3. Results and discussion

In our earlier studies [3,4], we demonstrated the performance of single phototransistor devices. Here, we focus on the device array for image sensing applications. Figure 1 illustrates the device architecture and operation mechanism. Electrons and holes generated in the depletion region of the Si substrate create a high field effect through the HfO_2 dielectric, which acts as a capacitor. The Fermi level of graphene is modulated by electrostatic doping, and this modulation is detected as a drain current in the readout circuit.

Broadband response is essential for image sensors. As shown in Fig. 2a, our device exhibits a broadband response across 400–900 nm when tested with incandescent and Xe lamps using color filters. The photoresponse (photocurrent) varies with wavelength and intensity, confirming its broadband capability.

To assess reproducibility and reliability, we fabricated a 10 × 10 device array (Fig. 1c-d). Figure 2b shows the current variations of 20 devices under dark conditions and 808 nm illumination at a 0.5 V bias, demonstrating consistent performance. Figures 3a-c display the current distributions of the array under 266 nm, 532 nm, and 808 nm illumination, respectively. Minor variations in photocurrent are attributed to the quality of transferred graphene, as direct synthesis of high-quality graphene on non-metal

substrates remains challenging. In contrast, high-quality $MoTe_2$ and other 2D transition-metal dichalcogenides (e.g., MoS_2 and $PtSe_2$) can be grown directly [4-7]. Additionally, Fig. 3d shows a photocurrent image mapping of the array forming the letters "ZJU" under RGB light illumination, further validating the broadband response. Higher-resolution images can be achieved by increasing data points and improving 2D material quality. These results underscore the potential of our device array for imaging applications.

4. Conclusions

We demonstrate that integrating 2D materials with high-*k* HfO_2 and conventional Si enables excellent broadband performance from UV to near-infrared wavelengths. The devices operate at low bias (e.g., 0.5 V source-drain voltage) and exhibit high responsivity, making them suitable for optical signal detection. Their simple structure and robust performance highlight their potential as image sensors for robotics and intelligent machine vision systems.

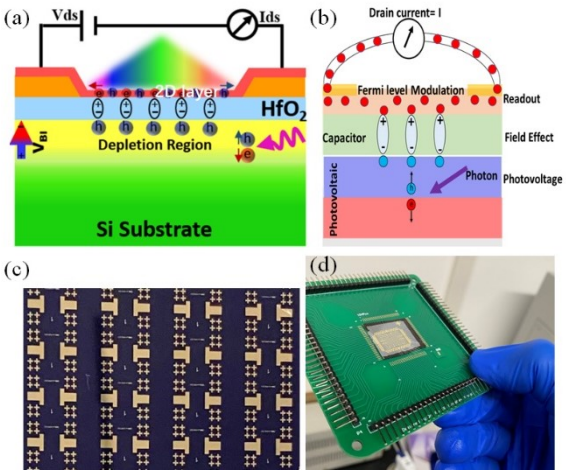

Fig. 1 (a) and (b) Phototransistor structure where the Si as a photovoltaic layer, the HfO_2 as a capacitive layer and the 2D layer as a channel for carrier transport, and its work mechanism; (c) and (d) 10 * 10 device array.

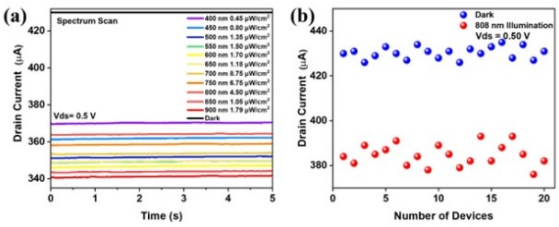

Fig. 2 (a) The spectrum scan of the device by the spectrometer (400 nm to 900 nm); (b) The repeatability and reliability test under the same measurement conductions (dark and 808 nm illumination) for 20 devices.

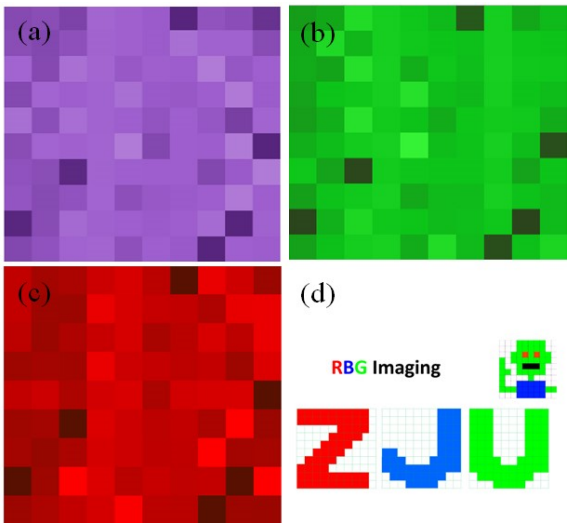

Fig. 3 (a)-(c) current mappings at 266 nm, 532 nm, and 808 nm illuminations, respectively; (d) The photocurrent image mapping of the 10 × 10 array of the shape of Z, J, and U under illumination of RBG light.

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